

NE 470/570 Monte Carlo Methods for Radiation Transport

Topic 5a: Derivation of the Linear Boltzmann Equation

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Outline

1. Introduction
2. Independent Variables
3. Basic Physics and Quantities
4. Internal and Boundary Sources
5. Time-dependent and Steady-state LBEs

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- This lecture is based on the book:

Anil K. Prinja, & Edward W. Larsen. (2010). Volume I Nuclear Engineering Fundamentals Chapter 5: General principles of neutron transport. In Handbook of Nuclear Engineering, edited by Dan G. Cacuci. Springer, ISBN: 978-0-387-98130-7.

<https://www.springer.com/gp/book/9780387981307>

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- This book chapter describes the basic theory underlying the neutron transport equation and the principal approximations used in this equation's applications to reactor physics.
 - 1 a discussion of the lack of smoothness of the angular flux in multi-dimensional geometries;
 - 2 derivations of the transport equation in specialized 1-D, 2-D, and 3-D geometries;
 - 3 a derivation of the time-dependent integral transport equation;
 - 4 an asymptotic derivation of the point kinetics equation;
 - 5 an asymptotic derivation of the multigroup P_1 and diffusion equations.

- A central problem in the design and analysis of nuclear reactors is the accurate and detailed prediction of the **space, angle, energy, and time-dependence of neutron distributions** in all components of the reactor.
- Neutrons are responsible for propagating the chain reaction and releasing energy through fission, but neutrons and photons, through fission, capture, scattering, and excitation/ionization interactions, also induce a thermal-mechanical response in the reactor core and, moreover, cause degradation of structural components, fuel rods, and the control system, through radiation damage, depletion, and fission product poison buildup.
- The transport of neutral radiation, including but not restricted to neutrons and photons, through matter is extremely well described by the **transport equation, a linear version of Boltzmann's equation** originally developed within the framework of the kinetic theory of gases (for this reason, **the transport equation is sometimes also referred to as the linear Boltzmann equation (LBE)**).
- This equation is an **integro-differential equation** having (generally) **seven independent variables**, whose solution is not smooth, and which can only be solved exactly for the simplest of problems. Essentially, all neutron transport problems of practical interest must be solved either approximately or numerically (deterministic and Monte Carlo).

- **Neutron transport** is the process in which neutrons propagate through the atoms in a physical system.
- This includes
 - 1 the **streaming** of neutrons from one collision site (an atomic nucleus) to the next;
 - 2 the **scattering** of neutrons off nuclei;
 - 3 the **capture** of neutrons by nuclei;
 - 4 the initiation by neutrons of **fission** events, in which a nucleus splits and two or more neutrons are emitted.
- In this lecture, we will develop the mathematical equations that describe the neutron transport process.
- The following topics will be included:
 - 1 the necessary independent variables
 - 2 the basic physics of neutron transport
 - 3 the relevant unknowns (angular flux, precursor densities - for problems with delayed neutrons)
 - 4 the internal and boundary sources
 - 5 time-dependent and steady-state LBE

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- To characterize a 3-D neutron transport process, **seven independent variables** are required:

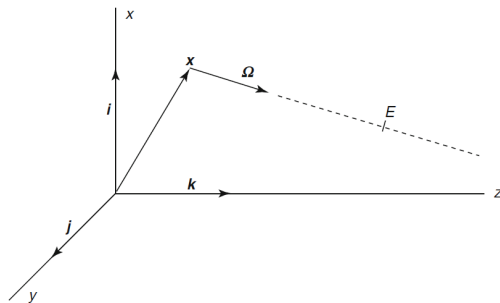
- 1 three components of the **position vector $\mathbf{x}$**
- 2 two angles to specify the unit vector $\mathbf{\Omega}$ denoting the **direction of flight**
- 3 the **kinetic energy E**
- 4 the **time t**

- These variables can specify the population of neutrons at

- 1 an arbitrary point $\mathbf{x}$ in the system
- 2 traveling in an arbitrary direction of flight $\mathbf{\Omega}$
- 3 with any energy E
- 4 at any time t

- For $\mathbf{x}$, we use the Cartesian coordinates

$$\mathbf{x} = x \cdot \mathbf{i} + y \cdot \mathbf{j} + z \cdot \mathbf{k} \quad (1)$$



- The direction-of-flight vector Ω , which has a **unit size** $||\Omega|| = 1$, is specified using a **polar angle** θ , defined relative to **the z -axis**, and an **azimuthal angle** ω , defined relative to **the x -axis**.

- In terms of the direction cosines projected onto the three Cartesian axes, we have:

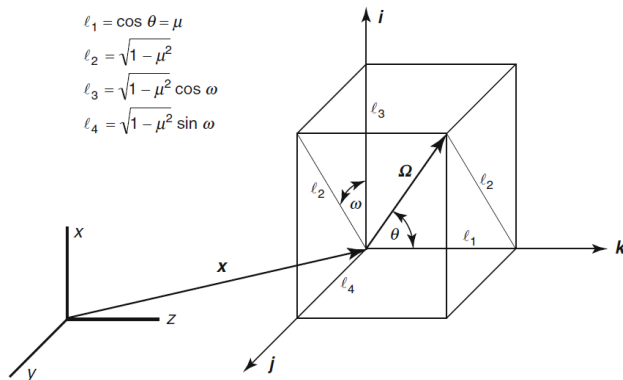
$$\Omega = \Omega_x \cdot \mathbf{i} + \Omega_y \cdot \mathbf{j} + \Omega_z \cdot \mathbf{k} \quad (2)$$

where

$$\Omega_x = l_3 = \sqrt{1 - \mu^2} \cos \omega$$

$$\Omega_y = l_4 = \sqrt{1 - \mu^2} \sin \omega$$

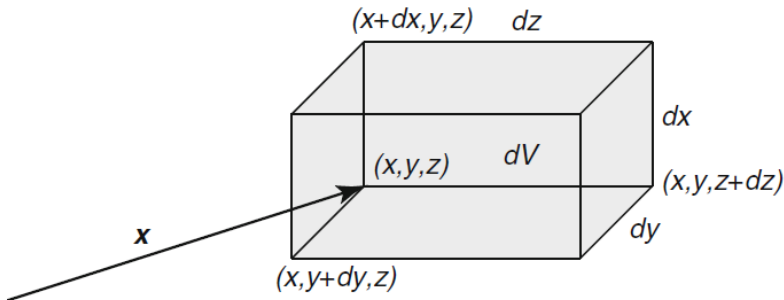
$$\Omega_z = l_1 = \mu = \cos \theta$$



Incremental volume in $\mathbf{x}$

- If we independently displace the spatial variables x , y , and z by incremental (very small) amounts dx , dy , and dz , the spatial variable $\mathbf{x}$ will sweep out an incremental hexahedral volume about $\mathbf{x}$.

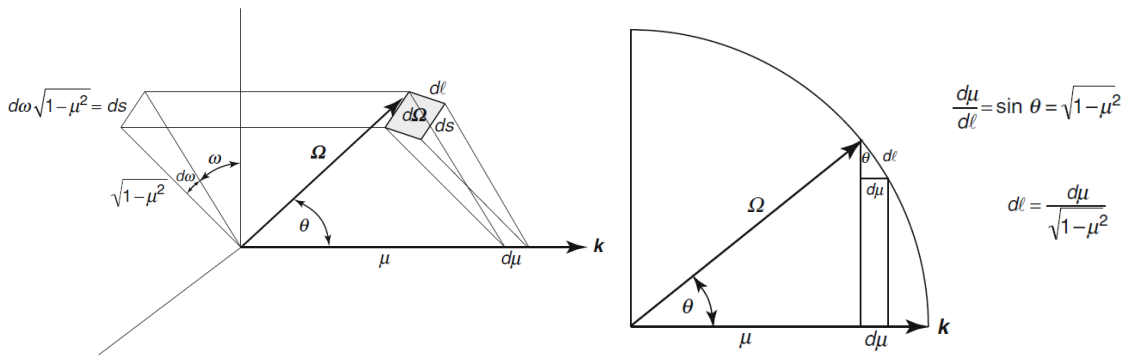
$$dV = dx \cdot dy \cdot dz \quad (3)$$



Incremental volume in Ω

- Similarly, if we independently displace the angular variables $\mu = \cos \theta$ and ω by incremental amounts $d\mu$ and $d\omega$, then the unit vector Ω will sweep out an incremental, rectangular, dimensionless element of area or **solid angle** on the unit sphere.

$$d\Omega = ds \cdot dl \quad (4)$$



- It is easy to get

$$ds = d\omega \cdot \sqrt{1 - \mu^2} \quad (5)$$

- Using the plane generated by Ω and k , we can get

$$dl = \frac{d\mu}{\sqrt{1 - \mu^2}} \quad (6)$$

- Thus, $d\Omega$ is given by:

$$d\Omega = ds \cdot dl = d\omega \cdot \sqrt{1 - \mu^2} \cdot \frac{d\mu}{\sqrt{1 - \mu^2}} = d\mu \cdot d\omega \quad (7)$$

- For example, the surface area of the entire unit sphere is then:

$$\int_{4\pi} d\Omega = \int_0^{2\pi} \int_{-1}^1 d\mu d\omega = 4\pi \quad (8)$$

where integration over the unit sphere is denoted by:

$$\int_{4\pi} f(\Omega) d\Omega = \int_0^{2\pi} \int_{-1}^1 f(\Omega) d\mu d\omega = \int_0^{2\pi} \int_{-1}^1 f(\mu, \omega) d\mu d\omega \quad (9)$$

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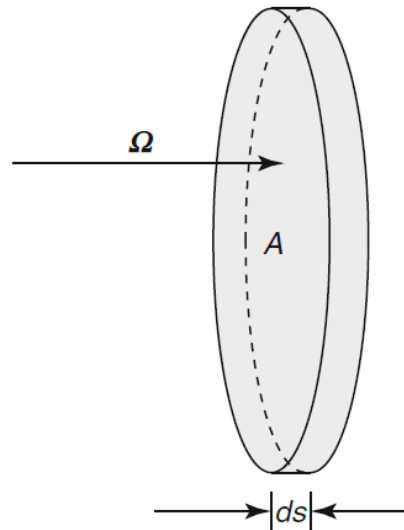
Outline

3. Basic Physics and Quantities

3.1. Basic Physics of Neutron Transport

3.2. Angular Neutron Density and Angular Flux

- Let us consider a neutron, streaming with direction Ω and energy E inside a material with known physical properties.
- As the neutron travels an **incremental distance ds** , there is an **incremental probability dp** that the neutron will interact with a nucleus.
- To determine the relationship between dp and ds , consider the neutron to be **normally incident**, at an arbitrary point, on a **target of area A** and **incremental thickness ds** .
- We assume that the **microscopic cross-sectional area of a target nucleus $\sigma(E)$ (cm^2)** and the **number density of target nuclei N (cm^{-3})** are known.



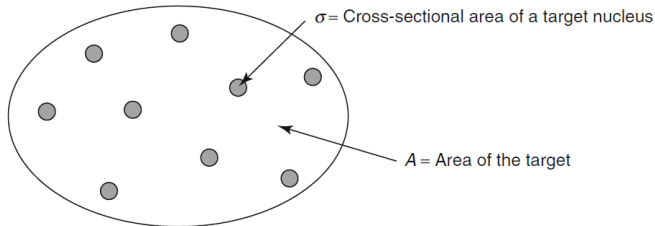
- The neutron's-eye-view of the target is shown below. If the target is sufficiently thin that one target nucleus does not shield another, then the probability of a collision is:

$$dp = \frac{\text{total area of nuclei}}{\text{area of target}} = \frac{n\sigma}{A} \quad (10)$$

where $n = N \cdot dV = N \cdot A \cdot ds$ is the **number of nuclei** in the target.

- We have

$$\begin{aligned} dp &= \frac{n\sigma}{A} \\ &= \frac{N \cdot A \cdot ds \cdot \sigma}{A} = N\sigma \cdot ds \end{aligned}$$



- Therefore, dp is proportional to ds :

$$dp = \Sigma_t(E)ds \quad (11)$$

where the “constant” of proportionality is the **total macroscopic cross section** (cm^{-1})

$$\Sigma_t(E) = N\sigma(E) \quad (12)$$

- When a neutron collides with a nucleus, it is **captured** with probability $p_c(E)$, **scatters** with probability $p_s(E)$, or initiates a **fission** event with probability $p_f(E)$. We ignore other possible, but generally rare events such as (n, α) , (n, p) , and $(n, 2n)$. Then we have:

$$p_c(E) + p_s(E) + p_f(E) = 1 \quad (13)$$

- Define $r = (c, s, f)$, the probabilities also define the macroscopic cross sections $\Sigma_r(E)$ such that

$$\Sigma_r(E)ds = \Sigma_t(E)ds \cdot p_r(E) \quad (14)$$

is the probability that a neutron with energy E , while traveling a distance ds , will experience a collision of type r . The total macroscopic cross section then satisfies:

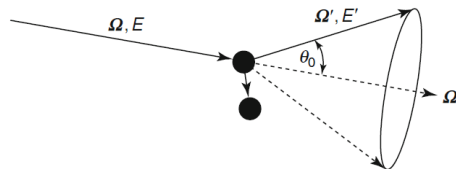
$$\Sigma_t(E) = \Sigma_c(E) + \Sigma_s(E) + \Sigma_f(E) \quad (15)$$

- If the neutron is **captured**, it is considered to be removed from the system. If the neutron (with direction Ω and energy E) **scatters**, it emerges from the scattering event with a new direction Ω' and energy E' . We assume that the distribution of post-collision directions Ω' and energies E' is known and can be expressed as:

$$p(\Omega \cdot \Omega', E \rightarrow E') d\Omega' dE' \quad (16)$$

which is formally defined as **the probability that the scattered neutron has direction in $d\Omega'$ about Ω' and energy in dE' about E' .**

- In writing p as $p(\Omega \cdot \Omega', E \rightarrow E') d\Omega' dE'$, we indicate that scattering in media with randomly distributed scattering centers (nuclei) is **rotationally invariant**. That is, the probability that a neutron will scatter from direction Ω to direction Ω' depends only on the scattering angle θ_0 between Ω and Ω' , or on **the cosine of this angle**, $\mu_0 = \cos \theta_0 = \Omega \cdot \Omega'$. Thus, all scattered directions of flight Ω' on the cone of **equal scattering angle** are **equally probable**.



- We now define the **macroscopic differential scattering cross section** as:

$$\Sigma_s(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}', E \rightarrow E') = \Sigma_s(E) \cdot p(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}', E \rightarrow E') \quad (17)$$

which has dimension $\text{cm}^{-1}\text{MeV}^{-1}$.

- Therefore,

$$\Sigma_s(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}', E \rightarrow E') ds d\boldsymbol{\Omega}' dE' \quad (18)$$

is the probability that a neutron, with direction $\boldsymbol{\Omega}$ and energy E and travelling a distance ds , will scatter into $d\boldsymbol{\Omega}'$ about $\boldsymbol{\Omega}'$ and dE' about E' .

- We have:

$$\begin{aligned} \int_0^\infty \int_{4\pi} p(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}', E \rightarrow E') d\boldsymbol{\Omega}' dE' &= 1 \\ \int_0^\infty \int_{4\pi} \Sigma_s(\boldsymbol{\Omega} \cdot \boldsymbol{\Omega}', E \rightarrow E') d\boldsymbol{\Omega}' dE' &= \Sigma_s(E) \end{aligned}$$

- If a neutron with energy E initiates a fission event, the target nucleus splits into two smaller **daughter nuclei**, and on the average, $\nu(E)$ neutrons are released.
- Of this number, $\nu(E)[1 - \beta(E)]$ are **prompt** (emitted within 10^{-15} s of the fission event) and $\nu(E)\beta(E)$ are **delayed** (emitted roughly 0.1 – 60 s after the fission event).
- Thus, the **delayed neutron fraction** $\beta(E)$ is the probability that a fission neutron, created by a neutron with energy E , is delayed ($\beta \approx 0.01$).
- Delayed fission neutrons are created from the **radioactive decay of unstable daughter nuclei**, which are produced during fission events.
- Prompt fission neutrons are born at the location of the fission event, their **initial direction of flight** is **isotropic** (uniformly distributed on the unit sphere), and their **initial energy** is consistent with the **prompt fission spectrum** $\chi_p(E)$, defined by:

$$\chi_p(E)dE \quad (19)$$

which is **the probability that a prompt fission neutron has energy in dE about E** . It satisfies

$$\int_0^\infty \chi_p(E)dE = 1 \quad (20)$$

- The unstable daughter nuclei are often grouped into **six precursor groups**, each with its own **radioactive decay constant** λ_j , **delayed neutron fraction** $\beta_j(E)$, and **delayed neutron fission spectrum** $\chi_j(E)$, $j = 1, 2, \dots, 6$.
- The functions $\beta_j(E)$ satisfy:

$$\sum_{j=1}^6 \beta_j(E) = \beta(E) \quad (21)$$

- We next introduce the notion of the neutron **mean free path (MFP)**.
- Consider a large number (N_0) of neutrons, all emitted in the same direction of flight, with the same energy E , and from the same spatial point within a large homogeneous region.
- Each neutron travels along the same flight path within the system and eventually undergoes a collision with a nucleus.
- Let $N(s, E)$ be the number of neutrons that penetrate to depth s without experiencing a collision.
- Clearly, $N(s, E)$ is a decreasing function of s , starting with the initial value

$$N(s = 0, E) = N_0$$

- For $s > 0$ and an incremental distance $ds > 0$, the number of neutron-nucleus collisions between s and $s + ds$ is given by:

$$N(s, E) - N(s + ds, E) \approx N(s, E) - \left(N(s, E) + ds \cdot \frac{\partial N(s, E)}{\partial s} \right) = -ds \cdot \frac{\partial N(s, E)}{\partial s} \quad (22)$$

- Dividing by $N(s, E)$, it becomes the probability that a neutron will have a collision while traveling a distance ds .
- Previously we have shown that $dp = \Sigma_t(E)ds$, so we have

$$-\frac{1}{N(s, E)} \cdot ds \cdot \frac{\partial N(s, E)}{\partial s} = \Sigma_t(E)ds \quad (23)$$

- This relationship can be rearranged to yield the differential equation:

$$\frac{\partial N(s, E)}{\partial s} + \Sigma_t(E) \cdot N(s, E) = 0, \quad N(s = 0, E) = N_0 \quad (24)$$

which has the solution:

$$N(s, E) = N_0 \cdot e^{-\Sigma_t(E) \cdot s} \quad (25)$$

- Next, define $p(s, E)ds$ as the probability that an uncollided neutron will collide with a nucleus between s and $s + ds$, we have:

$$\begin{aligned}
 p(s, E)ds &= \frac{\text{the number of collisions between } s \text{ and } s + ds}{\text{the initial number of neutrons (at } s = 0)} \\
 &= \frac{1}{N_0} \left(-\frac{\partial N(s, E)}{\partial s} \cdot ds \right) = \frac{1}{N_0} \left(N_0 \cdot \Sigma_t(E) \cdot e^{-\Sigma_t(E) \cdot s} \cdot ds \right) \\
 &= \Sigma_t(E) \cdot e^{-\Sigma_t(E) \cdot s} ds
 \end{aligned}$$

- So we have:

$$p(s, E) = \Sigma_t(E) \cdot e^{-\Sigma_t(E) \cdot s} \quad (26)$$

which satisfies the normalization property

$$\int_0^\infty p(s, E)ds = \int_0^\infty \Sigma_t(E) \cdot e^{-\Sigma_t(E) \cdot s} ds = 1 \quad (27)$$

- The probability density $p(s, E)$ can be used to obtain the mean distance-to-collision, or the MFP:

$$\lambda(E) = \int_0^\infty s p(s, E)ds = \int_0^\infty s \Sigma_t(E) \cdot e^{-\Sigma_t(E) \cdot s} ds = \frac{1}{\Sigma_t(E)} \quad (28)$$

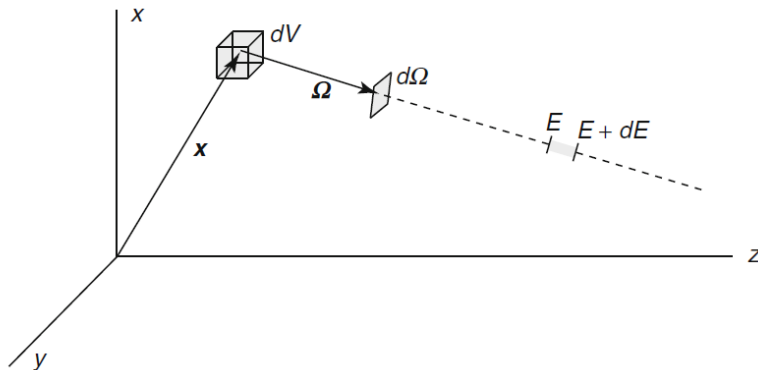
Outline

3. Basic Physics and Quantities

3.1. Basic Physics of Neutron Transport

3.2. Angular Neutron Density and Angular Flux

- We now define the physical quantities needed to describe the neutron population.
- We consider all neutrons that, at time t
 - 1 are geometrically located in a volume increment dV about a point $\mathbf{x}$,
 - 2 travel within a **solid angle** $d\Omega$ about the direction Ω ,
 - 3 have energies between E and $E + dE$



The angular neutron density $N(\mathbf{x}, \boldsymbol{\Omega}, E, t)$

- Define the **angular neutron density** $N(\mathbf{x}, \boldsymbol{\Omega}, E, t)$, a function of (generally) **seven** independent variables, such that:

$$N(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE \quad (29)$$

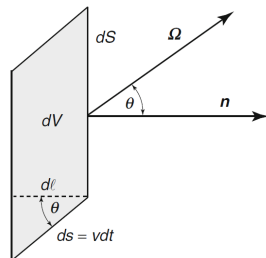
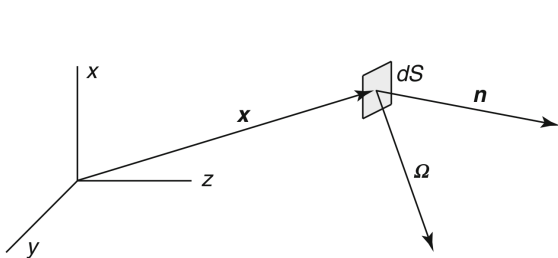
physically means the **number of neutrons in $dV d\boldsymbol{\Omega} dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at time t .**

- This definition implies that the value of $N(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ is independent of the increments dV , $d\boldsymbol{\Omega}$, and dE .
- Its unit is **$\text{cm}^{-3} \text{MeV}^{-1}$** . Also, $N(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ is a phase space density.
- Knowledge of $N(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ enables one to calculate the number of neutrons that exist in any “volume” of 6-D space $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at any time t .
- For example, for any $E_1 < E_2$ and any subregion R of the physical system V , we have:

$$\int_{E_1}^{E_2} \int_{4\pi} \int_R N(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE = n(E) \quad (30)$$

which physically means the **the number of neutrons in R , with energies between E_1 and E_2 , at time t .**

- Next, let us consider a planar surface area element dS , located at any point $\mathbf{x}$ in V , with a **unit normal vector $\mathbf{n}$** .
- We consider neutrons located near $\mathbf{x}$, traveling in directions in $d\Omega$ about Ω , with energies in dE about E , at time t .
- What is **the rate (number per second) at which these neutrons flow through dS at time t** ?
- In the plane generated by Ω and $\mathbf{n}$, and with $v = \sqrt{\frac{2E}{m}}$ as the neutron speed, we look at a volume dV obtained by sweeping dS a distance $ds = vdt$ along the direction of flight Ω .



$$\frac{d\ell}{ds} = \cos \theta = \Omega \cdot \mathbf{n}$$

$$d\ell = ds (\Omega \cdot \mathbf{n}) = vdt(\Omega \cdot \mathbf{n})$$

- In time increment dt , neutrons travel the distance $ds = vdt$. Thus, in time dt , among all neutrons traveling in $d\Omega$ about Ω and dE about E , only those physically located in the incremental volume dV will flow through dS .

- The volume¹ of dV is:

$$dV = dS \cdot dl = dS \cdot v \cdot dt \cdot (\Omega \cdot \mathbf{n}) \quad (31)$$

- Thus, the number of neutrons located within $dVd\Omega dE$ about $(\mathbf{x}, \Omega, E)$ is

$$NdVd\Omega dE = N \cdot dS \cdot v \cdot dt \cdot (\Omega \cdot \mathbf{n})d\Omega dE = (\Omega \cdot \mathbf{n})vNdSd\Omega dEdt \quad (32)$$

which physically also means the number of neutrons traveling in $d\Omega dE$ about (Ω, E) that pass through dS during time dt about t .

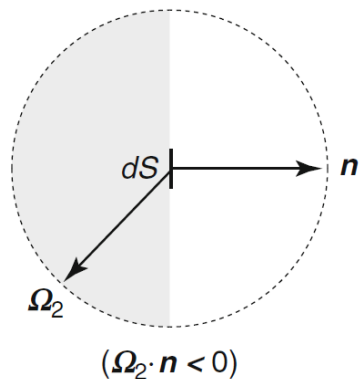
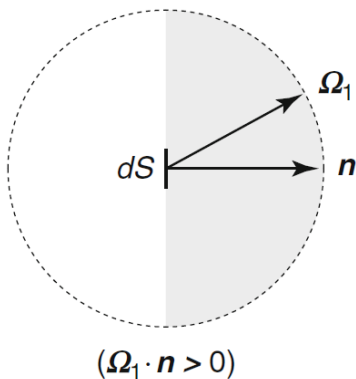
- Dividing by dt , we obtain:

$$(\Omega \cdot \mathbf{n})vN(\mathbf{x}, \Omega, E, t)dSd\Omega dE \quad (33)$$

as the rate (number per second) at which neutrons, traveling in $d\Omega dE$ about (Ω, E) , flow through dS at time t .

¹Unfortunately, on the previous slide, the figure on the right does not show a control volume properly.

- Each incremental surface dS has two normal vectors, $\mathbf{n}$ and $-\mathbf{n}$. The choice of $\mathbf{n}$ determines “positive” and “negative” directions of flow through dS .
- A positive (negative) rate indicates flow through dS in the hemisphere of directions $\boldsymbol{\Omega} \cdot \mathbf{n} > 0$ ($\boldsymbol{\Omega} \cdot \mathbf{n} < 0$).



The angular flux $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$

- The **angular flux (or fluence rate)** $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ is defined by:

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) = vN(\mathbf{x}, \boldsymbol{\Omega}, E, t) \quad (34)$$

where $v = \sqrt{\frac{2E}{m}}$ is the the neutron speed (m/s).

- The unit of $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ is $\text{cm}^{-2}\text{MeV}^{-1}\text{s}^{-1}$.
- A physical interpretation for ψ is obtained by considering neutrons in $dVd\boldsymbol{\Omega}dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at time t . During a time increment dt about t

$$\psi dVd\boldsymbol{\Omega}dEdt = vNdVd\boldsymbol{\Omega}dEdt = (vdt) \cdot (NdVd\boldsymbol{\Omega}dE) \quad (35)$$

which physically means the “**distance (path length) traveled by one neutron in time dt** ” multiplied by “**number of neutrons in $dVd\boldsymbol{\Omega}dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$** ”.

- Or equivalently, this means **the total path length traveled by neutrons, in $dVd\boldsymbol{\Omega}dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$, during time increment dt about t .**

- Dividing by dt , we obtain a **volume-based interpretation** for ψ :

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE \quad (36)$$

is the rate at which path length is generated by neutrons in $dV d\boldsymbol{\Omega} dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at time t .

- Hence, ψ is sometimes called the **path length density**.
- Finally, we can have a **surface-based interpretation** for ψ :

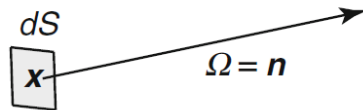
$$|\boldsymbol{\Omega} \cdot \mathbf{n}| \psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) dS d\boldsymbol{\Omega} dE \quad (37)$$

is the absolute rate that neutrons, traveling in $d\boldsymbol{\Omega} dE$ about $(\boldsymbol{\Omega}, E)$, flow through dS at time t .

- If the surface increment dS is perpendicular to the direction of neutron travel, i.e., if $\mathbf{n} = \boldsymbol{\Omega}$, we have $|\boldsymbol{\Omega} \cdot \mathbf{n}| = 1$.

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) dS d\boldsymbol{\Omega} dE \quad (38)$$

is the absolute rate at which neutrons, traveling in $d\boldsymbol{\Omega} dE$ about $(\boldsymbol{\Omega}, E)$, flow through a surface increment dS perpendicular to $\boldsymbol{\Omega}$ at time t .



- Equation (36) defines ψ in terms of a volume element dV , while Equations (37) and (38) define ψ in terms of a surface element dS . Both interpretations are useful.
- For example, if R is a subregion of V and $\mathbf{n}$ is the unit outer normal vector on ∂R , then Equation (38) implies that

$$J_R^+(t) = \int_0^\infty \int_{\partial R} \int_{\mathbf{\Omega} \cdot \mathbf{n} > 0} (\mathbf{\Omega} \cdot \mathbf{n}) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) dS d\mathbf{\Omega} dE \quad (39)$$

physically means the rate at which neutrons flow (leak) out of R at time t , and

$$J_R^-(t) = \int_0^\infty \int_{\partial R} \int_{\mathbf{\Omega} \cdot \mathbf{n} < 0} (|\mathbf{\Omega} \cdot \mathbf{n}|) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) dS d\mathbf{\Omega} dE \quad (40)$$

physically means the rate at which neutrons flow into R at time t .

- Therefore,

$$J_R(t) = J_R^+(t) - J_R^-(t) = \int_0^\infty \int_{\partial R} \int_{4\pi} (\mathbf{\Omega} \cdot \mathbf{n}) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) dS d\mathbf{\Omega} dE \quad (41)$$

physically means the net rate at which neutrons flow (leak) out of R at time t .

- Using the **divergence theorem**,

$$\int_{\partial R} (\mathbf{n} \cdot \mathbf{f}(\mathbf{x})) dS = \int_R (\nabla \cdot \mathbf{f}(\mathbf{x})) dV \quad (42)$$

- Equation (41) becomes:

$$J_R(t) = J_R^+(t) - J_R^-(t) = \int_0^\infty \int_R \int_{4\pi} (\boldsymbol{\Omega} \cdot \nabla) \psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE \quad (43)$$

- For the surface definition in Equation (37), it can be used to determine the rates at which neutrons flow across surfaces. Equations (39) and (40) also hold for surfaces that are not closed, i.e., that do not enclose a volume.
- For the volume definition in Equation (36), it is useful because the number of collisions with nuclei that neutrons experience in an incremental time interval is proportional to the incremental distance that the neutrons travel during that time interval.
- Thus, **the rate at which neutrons interact with nuclei in a volume is proportional to the rate at which the neutrons generate path length in the volume**. The constants of proportionality are the **macroscopic cross sections**.

- In addition to the neutron angular flux $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$, the precursor densities $C_j(\mathbf{x}, t)$ for the six delayed neutron precursor groups are also required.
- For $1 \leq j \leq 6$, and for an incremental volume dV about $\mathbf{x}$,

$$C_j(\mathbf{x}, t)dV \quad (44)$$

physically means the number of group j precursor nuclei in dV about $\mathbf{x}$ at time t .

- Each precursor nucleus will eventually decay and emit one delayed neutron.
- Thus, the rate at which group j precursor nuclei experience radioactive decay equals the rate at which delayed neutrons are emitted by these nuclei.

Outline

1. Introduction
2. Independent Variables
3. Basic Physics and Quantities
- 4. Internal and Boundary Sources**
5. Time-dependent and Steady-state LBEs

- An **internal neutron source** $Q(\mathbf{x}, E, t)$, generally produced by radioactive decay, is located inside the physical system V and is usually **isotropic** (in radioactive decay, neutrons are emitted in all directions Ω with equal probability).
- Q is defined such that

$$\frac{Q(\mathbf{x}, E, t)}{4\pi} dV d\Omega dE \quad (45)$$

physically means the incremental rate at which neutrons are produced in $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E)$ from an internal source, at time t .

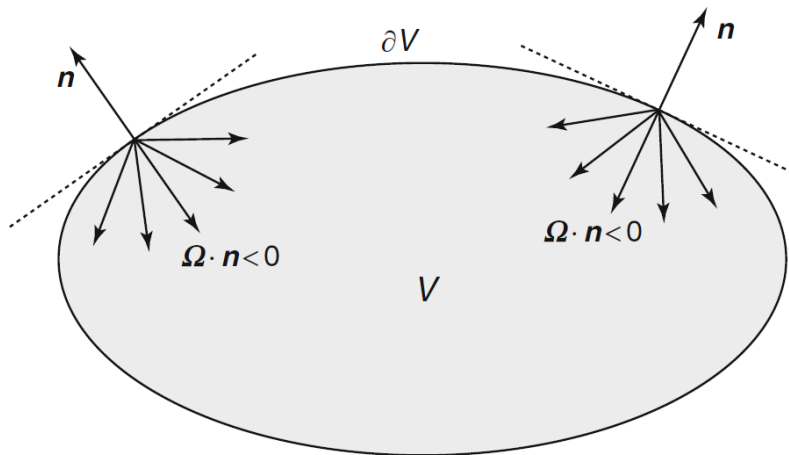
- This definition implies that the numerical value of Q is independent of the increments dV , $d\Omega$, and dE . The **factor** 4π is included so that integration of (45) over the unit sphere gives the following equivalent (for an isotropic source) definition:

$$Q(\mathbf{x}, E, t) dV dE \quad (46)$$

physically means the incremental rate at which neutrons are isotropically introduced into $dV dE$ about $(\mathbf{x}, E)$ from an internal source, at time t .

- $Q(\mathbf{x}, E, t)$ must be specified for all points $\mathbf{x}$ in V , all energies E , and all times after the initial time.

- **Boundary sources** are specified neutron fluxes that enter the physical system V through its outer boundary ∂V . V is usually assumed to be convex; then neutrons leaking out of V cannot reenter through ∂V .



- The **boundary angular flux** $\psi^b(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ is an external source, independent of the flux within the system, which must be specified for:
 - 1 all points on the outer boundary of the system, $\mathbf{x} \in \partial V$,
 - 2 all directions of flight pointing into the system, $\boldsymbol{\Omega} \cdot \mathbf{n} < 0$, where $\mathbf{n}$ is the unit outer normal vector at $\mathbf{x} \in \partial V$,
 - 3 all energies,
 - 4 all times after the initial time.
- The angular flux is required to satisfy:

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) = \psi^b(\mathbf{x}, \boldsymbol{\Omega}, E, t), \quad \mathbf{x} \in \partial V, \quad \boldsymbol{\Omega} \cdot \mathbf{n} < 0, \quad 0 < E < \infty, \quad t > 0 \quad (47)$$

- In the case of $\psi^b(\mathbf{x}, \boldsymbol{\Omega}, E, t) = 0$, the boundary ∂V is called a **vacuum boundary**.

- The internal and boundary neutron sources are specified and independent of the neutron distribution within V .
- Generally, neutron transport problems are driven by these known sources, which generate the “original” neutrons in the problem.
- After neutrons are introduced into V by the internal or boundary sources, they “transport” within V according to the physical processes of streaming, absorption, scattering, and fission, discussed above.
- In nuclear reactor problems, the term fission source describes the neutrons that are produced by fission events in V . This “source” directly depends on ψ and must be calculated.
- Likewise, the scattering source of neutrons emitted from scattering events also directly depends on ψ and must be calculated. (Thus, the term “source” sometimes describes a quantity that depends on ψ .) However, to repeat, the internal and boundary sources are independent of ψ and must be specified.

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1. Introduction
2. Independent Variables
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5. Time-dependent and Steady-state LBEs

5.1. The Time-Dependent Equations of Neutron Transport

5.2. The Time-dependent LBE Without Delayed Neutrons

5.3. The Steady-state LBE

5.4. Some different forms of the LBE

- We will **derive seven mathematical equations** that determine the neutron angular flux $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ and the six neutron precursor densities $C_j(\mathbf{x}, t)$.
- Each of these seven equations is a **conservation equation** - each is based on the simple physical concept that the rate of change (of neutrons and precursor densities within increments of phase space) equals the rate of gain minus the rate of loss.
- Consider the incremental population of neutrons within $dV d\boldsymbol{\Omega} dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at time t

$$dN = N(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE = \frac{1}{v} \psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) dV d\boldsymbol{\Omega} dE \quad (48)$$

- Thus, the time rate of change of dN is:

$$\frac{\partial N}{\partial t} = \frac{1}{v} \frac{\partial \psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)}{\partial t} dV d\boldsymbol{\Omega} dE = (\text{rate of gain}) - (\text{rate of loss}) \quad (49)$$

where

rate of loss = collision rate + net leakage rate

rate of gain = in-scattering rate + prompt fission rate + delayed fission rate + source rate

in $dV d\boldsymbol{\Omega} dE$ about $(\mathbf{x}, \boldsymbol{\Omega}, E)$ at time t .

- We first consider the **loss** terms. Using $v = \frac{ds}{dt}$, for neutrons in $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E)$, the **collision rate** is

$$\begin{aligned}\Sigma_t(E) \cdot \psi(\mathbf{x}, \Omega, E, t) dV d\Omega dE &= \Sigma_t(E) \frac{ds}{dt} N(\mathbf{x}, \Omega, E, t) dV d\Omega dE \\ &= \frac{1}{dt} [\Sigma_t(E) ds] \cdot [N(\mathbf{x}, \Omega, E, t) dV d\Omega dE]\end{aligned}\quad (50)$$

- The term $\Sigma_t(E) ds$ means the probability that a (single) neutron with energy E will undergo a collision in time interval dt . Also, the term $N(\mathbf{x}, \Omega, E, t) dV d\Omega dE$ means the number of neutrons in $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E, t)$ at time t . Their product means the number of collisions in $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E)$ in time interval dt at time t . After taking the derivative w.r.t time t , we obtain the collision rate.
- Based on Equation (41), we let R shrink to an incremental volume dV and directly get the **net leakage rate** as:

$$(\Omega \cdot \nabla) \psi(\mathbf{x}, \Omega, E, t) dV d\Omega dE \quad (51)$$

- Therefore, the **rate of loss** becomes:

$$\text{rate of loss} = [\Sigma_t(E) \cdot \psi(\mathbf{x}, \Omega, E, t) + (\Omega \cdot \nabla) \psi(\mathbf{x}, \Omega, E, t)] dV d\Omega dE \quad (52)$$

- Next we consider the **gain** terms. For neutrons in $dV d\Omega' dE'$ about $(\mathbf{x}, \Omega', E')$:

$$\begin{aligned}
 & \Sigma_s (\Omega' \cdot \Omega, E' \rightarrow E) \cdot \psi(\mathbf{x}, \Omega', E', t) dV d\Omega' dE' d\Omega dE \\
 &= \Sigma_s (\Omega' \cdot \Omega, E' \rightarrow E) \cdot \frac{ds}{dt} \cdot N(\mathbf{x}, \Omega', E', t) dV d\Omega' dE' d\Omega dE \\
 &= \frac{1}{dt} [N(\mathbf{x}, \Omega', E', t) dV d\Omega' dE'] \cdot [\Sigma_s (\Omega' \cdot \Omega, E' \rightarrow E) ds d\Omega dE]
 \end{aligned} \tag{53}$$

- The term $[N(\mathbf{x}, \Omega', E', t) dV d\Omega' dE']$ means the number of neutrons in $dV d\Omega' dE'$ about $(\mathbf{x}, \Omega', E')$ at time t .
- The term $[\Sigma_s (\Omega' \cdot \Omega, E' \rightarrow E) ds d\Omega dE]$ means the probability that one of these neutrons will scatter into $d\Omega dE$ about (Ω, E) during time interval dt .
- Their product means the the number of neutrons in dV about $\mathbf{x}$ that scatter from $d\Omega' dE'$ about (Ω', E') into $d\Omega dE$ about (Ω, E) during time interval dt about time t .
- Divided by dt , it means the rate at which neutrons in dV about $\mathbf{x}$ scatter from $d\Omega' dE'$ about (Ω', E') into $d\Omega dE$ about (Ω, E) at time t .
- Therefore, the in-scattering rate can be derived based on Equation (53).

- Integrating this expression over all initial directions Ω' and energies E' , we obtain:

$$\left[\int_0^\infty \int_{4\pi} \Sigma_s(\Omega' \cdot \Omega, E' \rightarrow E) \cdot \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \right] dV d\Omega dE \quad (54)$$

as the rate at which neutrons in dV about $\mathbf{x}$ scatter into $d\Omega dE$ about (Ω, E) at time t . In other words, it is the **in-scattering rate**.

- Similarly, the rate at which prompt fission neutrons are produced in $dV d\Omega' dE'$ about $(\mathbf{x}, \Omega', E')$ is given by:

$$[1 - \beta(E')] \nu(E') \Sigma_f(E') \psi(\mathbf{x}, \Omega', E', t) dV d\Omega' dE' \quad (55)$$

- Integrating this expression over all initial directions Ω' and energies E' , and multiplying by $\frac{\chi_p(E)}{4\pi} d\Omega dE$, we obtain:

$$\left[\frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} [1 - \beta(E')] \nu(E') \Sigma_f(E') \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \right] dV d\Omega dE \quad (56)$$

as the rate at which prompt fission neutrons are produced in $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E)$ at time t . In other words, the **prompt fission rate**.

- Also, for the delayed neutrons:

$$\sum_{j=1}^6 \frac{\chi_j(E)}{4\pi} \lambda_j C_j(\mathbf{x}, t) dV d\Omega dE \quad (57)$$

means the rate at which delayed neutrons are emitted into $dV d\Omega dE$ about $(\mathbf{x}, \Omega, E)$ at time t . In other words, the **delayed fission rate**.

- Finally, by Equation (45), we have the **source rate** as:

$$\frac{Q(\mathbf{x}, E, t)}{4\pi} dV d\Omega dE \quad (58)$$

- Finally, the **rate of gain** becomes:

$$\begin{aligned} \text{rate of gain} = & \left[\int_0^\infty \int_{4\pi} \Sigma_s(\Omega' \cdot \Omega, E' \rightarrow E) \cdot \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \right. \\ & + \frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} [1 - \beta(E')] \nu \Sigma_f(E') \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \\ & \left. + \sum_{j=1}^6 \frac{\chi_j(E)}{4\pi} \lambda_j C_j(\mathbf{x}, t) + \frac{Q(\mathbf{x}, E, t)}{4\pi} \right] dV d\Omega dE \end{aligned} \quad (59)$$

Note that we have removed the dependence of ν on E .

- Put the rate of gain and rate of loss into the conservation equation:

$$\begin{aligned} \frac{1}{v} \frac{\partial \psi}{\partial t} dV d\mathbf{\Omega} dE = & \left[\int_0^\infty \int_{4\pi} \Sigma_s \psi d\mathbf{\Omega}' dE' + \frac{\chi_p}{4\pi} \int_0^\infty \int_{4\pi} [1 - \beta] \nu \Sigma_f \psi d\mathbf{\Omega}' dE' \right. \\ & \left. + \frac{1}{4\pi} \sum_{j=1}^6 \chi_j \lambda_j C_j + \frac{1}{4\pi} Q \right] dV d\mathbf{\Omega} dE - [\Sigma_t \psi + (\mathbf{\Omega} \cdot \nabla) \psi] dV d\mathbf{\Omega} dE \end{aligned} \quad (60)$$

- Dividing by $dV d\mathbf{\Omega} dE$ and rearranging, we obtain the following first-order integro-differential equation:

$$\begin{aligned} \frac{1}{v} \frac{\partial \psi(\mathbf{x}, \mathbf{\Omega}, E, t)}{\partial t} + (\mathbf{\Omega} \cdot \nabla) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) + \Sigma_t(E) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) \\ = \int_0^\infty \int_{4\pi} \Sigma_s(\mathbf{\Omega}' \cdot \mathbf{\Omega}, E' \rightarrow E) \cdot \psi(\mathbf{x}, \mathbf{\Omega}', E', t) d\mathbf{\Omega}' dE' \\ + \frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} [1 - \beta(E')] \nu \Sigma_f(E') \psi(\mathbf{x}, \mathbf{\Omega}', E', t) d\mathbf{\Omega}' dE' \\ + \frac{1}{4\pi} \sum_{j=1}^6 \chi_j(E) \lambda_j C_j(\mathbf{x}, t) + \frac{Q(\mathbf{x}, E, t)}{4\pi} \end{aligned} \quad (61)$$

- This is the **time-dependent neutron transport equation** (with delayed neutron precursors), also called the **linear Boltzmann equation**.

- We must also derive an equation for each precursor density. The **time rate of change** of the group j precursor density may be expressed as:

$$\frac{\partial C_j(\mathbf{x}, t)}{\partial t} dV = (\text{rate of gain}) - (\text{rate of loss}) \quad (62)$$

which physically means the rate of change of the number of group j precursor nuclei in dV about $\mathbf{x}$ at time t .

- The **rate of loss** (due to decay) of these precursors is given by:

$$\lambda_j C_j(\mathbf{x}, t) dV \quad (63)$$

which is the rate at which group j precursor nuclei undergo decay in dV about $\mathbf{x}$ at time t .

- The **rate of gain** is given by:

$$\beta_j(E') \nu \Sigma_f(E') \psi(\mathbf{x}, \boldsymbol{\Omega}', E', t) dV d\boldsymbol{\Omega}' dE' \quad (64)$$

which is the rate at which group j precursor nuclei are produced by neutrons in $dV d\boldsymbol{\Omega}' dE'$ about $(\mathbf{x}, \boldsymbol{\Omega}', E')$ at time t .

- Integrate Equation (64) over Ω' and E' , we get the total rate of gain of the group j precursor nuclei:

$$\left[\int_0^\infty \int_{4\pi} \beta_j(E') \nu \Sigma_f(E') \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \right] dV \quad (65)$$

which physically means the rate at which group j precursor nuclei are produced in dV about $\mathbf{x}$ at time t .

- By combining Equations (62), (63) and (65), dividing both sides by dV and re-arrange, we obtain:

$$\frac{\partial C_j(\mathbf{x}, t)}{\partial t} + \lambda_j C_j(\mathbf{x}, t) = \int_0^\infty \int_{4\pi} \beta_j(E') \nu \Sigma_f(E') \psi(\mathbf{x}, \Omega', E', t) d\Omega' dE' \quad (66)$$

for each group j , $1 \leq j \leq 6$.

- As with the transport equation, each term in these equations describes a physical process that causes a gain or loss of group j delayed neutron precursor nuclei.

- Equation (66) must be solved jointly with Equation (61) - the neutron transport equation.
- In addition, ψ must satisfy the boundary conditions:

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t) = \psi^b(\mathbf{x}, \boldsymbol{\Omega}, E, t), \quad \mathbf{x} \in \partial V, \quad \boldsymbol{\Omega} \cdot \mathbf{n} < 0, \quad 0 < E < \infty, \quad t > 0 \quad (67)$$

where ψ^b is given.

- ψ and c_j must satisfy the initial conditions:

$$\begin{aligned} \psi(\mathbf{x}, \boldsymbol{\Omega}, E, t = 0) &= \psi^i(\mathbf{x}, \boldsymbol{\Omega}, E), \quad \mathbf{x} \in V, \quad \boldsymbol{\Omega} \in 4\pi, \quad 0 < E < \infty \\ C_j(\mathbf{x}, t = 0) &= C_j^i(\mathbf{x}), \quad \mathbf{x} \in V \end{aligned} \quad (68)$$

where ψ^i and ∂C_j^i are given.

- In full generality, the neutron angular flux $\psi(\mathbf{x}, \boldsymbol{\Omega}, E, t)$ and the precursor densities $C_j(\mathbf{x}, t)$ are obtained by solving the seven equations, Equations (61) and (66), subject to the boundary condition stated in Equation (67) and the initial conditions stated in Equation (68).
- The subject of nuclear reactor kinetics is based on this system of equations.

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5.1. The Time-Dependent Equations of Neutron Transport

5.2. The Time-dependent LBE Without Delayed Neutrons

5.3. The Steady-state LBE

5.4. Some different forms of the LBE

- In some time-dependent problems, the relatively small value of $\beta \approx 0.01$ justifies the neglect of the delayed neutron terms.
- In such problems one can set $\beta_j = C_j = 0$. The resulting single transport equation for $\psi(\mathbf{x}, \mathbf{\Omega}, E, t)$ becomes:

$$\begin{aligned}
 \frac{1}{v} \frac{\partial \psi(\mathbf{x}, \mathbf{\Omega}, E, t)}{\partial t} + (\mathbf{\Omega} \cdot \nabla) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) + \Sigma_t(E) \psi(\mathbf{x}, \mathbf{\Omega}, E, t) \\
 = \int_0^\infty \int_{4\pi} \Sigma_s(\mathbf{\Omega}' \cdot \mathbf{\Omega}, E' \rightarrow E) \cdot \psi(\mathbf{x}, \mathbf{\Omega}', E', t) d\mathbf{\Omega}' dE' \\
 + \frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} \nu \Sigma_f(E') \psi(\mathbf{x}, \mathbf{\Omega}', E', t) d\mathbf{\Omega}' dE' + \frac{Q(\mathbf{x}, E, t)}{4\pi}
 \end{aligned} \tag{69}$$

- The boundary and initial conditions become:

$$\begin{aligned}
 \psi(\mathbf{x}, \mathbf{\Omega}, E, t) &= \psi^b(\mathbf{x}, \mathbf{\Omega}, E, t), \quad \mathbf{x} \in \partial V, \quad \mathbf{\Omega} \cdot \mathbf{n} < 0, \quad 0 < E < \infty, \quad t > 0 \\
 \psi(\mathbf{x}, \mathbf{\Omega}, E, 0) &= \psi^i(\mathbf{x}, \mathbf{\Omega}, E), \quad \mathbf{x} \in V, \quad \mathbf{\Omega} \in 4\pi, \quad 0 < E < \infty
 \end{aligned} \tag{70}$$

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- In steady-state problems, the precursor densities can be eliminated to obtain a single equation for ψ . Setting $\frac{\partial C_j(\mathbf{x}, t)}{\partial t} = 0$ in Equation (66), we get:

$$\lambda_j C_j(\mathbf{x}) = \int_0^\infty \int_{4\pi} \beta_j(E') \nu \Sigma_f(E') \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' \quad (71)$$

- Introduce this expression into the steady-state version of Equation (61)

$$\begin{aligned} & (\boldsymbol{\Omega} \cdot \nabla) \psi(\mathbf{x}, \boldsymbol{\Omega}, E) + \Sigma_t(E) \psi(\mathbf{x}, \boldsymbol{\Omega}, E) \\ &= \int_0^\infty \int_{4\pi} \Sigma_s(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}, E' \rightarrow E) \cdot \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' \\ &+ \frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} [1 - \beta(E')] \nu \Sigma_f(E') \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' \\ &+ \frac{1}{4\pi} \sum_{j=1}^6 \chi_j(E) \int_0^\infty \int_{4\pi} \beta_j(E') \nu \Sigma_f(E') \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' + \frac{Q(\mathbf{x}, E)}{4\pi} \end{aligned} \quad (72)$$

- Since $\beta = \sum_{j=1}^6 \beta_j \approx 0.01$, it is often acceptable to set $\beta_j = 0$. After ignoring the delayed neutrons, Equation (72) becomes

$$\begin{aligned}
 (\boldsymbol{\Omega} \cdot \nabla) \psi(\mathbf{x}, \boldsymbol{\Omega}, E) + \Sigma_t(E) \psi(\mathbf{x}, \boldsymbol{\Omega}, E) \\
 = \int_0^\infty \int_{4\pi} \Sigma_s(\boldsymbol{\Omega}' \cdot \boldsymbol{\Omega}, E' \rightarrow E) \cdot \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' \\
 + \frac{\chi_p(E)}{4\pi} \int_0^\infty \int_{4\pi} \nu \Sigma_f(E') \psi(\mathbf{x}, \boldsymbol{\Omega}', E') d\boldsymbol{\Omega}' dE' + \frac{Q(\mathbf{x}, E)}{4\pi}
 \end{aligned} \tag{73}$$

- The first term on the right side of this equation is often called the **scattering source**.
- The second term is the **fission source**; this is isotropic and separable in space and energy - it is the product of a (known) function of energy and an (unknown) function only of space.
- The third term is the specified **internal source**.
- The boundary and initial conditions become:

$$\psi(\mathbf{x}, \boldsymbol{\Omega}, E) = \psi^b(\mathbf{x}, \boldsymbol{\Omega}, E), \quad \mathbf{x} \in \partial V, \quad \boldsymbol{\Omega} \cdot \mathbf{n} < 0, \quad 0 < E < \infty \tag{74}$$

- As before, if the specified incident boundary flux $\psi^b = 0$, then ∂V is called a **vacuum boundary**.

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- $\mathbf{r}$ is used to denote the position vector

$$\mathbf{r} = x \cdot \mathbf{i} + y \cdot \mathbf{j} + z \cdot \mathbf{k} \quad (75)$$

so $\psi(\mathbf{x}, \Omega, E, t)$ becomes $\psi(\mathbf{r}, \Omega, E, t)$, dV becomes d^3r .

- $\hat{\Omega}$ is used instead of Ω , $\psi(\mathbf{r}, \Omega, E, t) \rightarrow \psi(\mathbf{r}, \hat{\Omega}, E, t)$.
- $\Sigma_s(\Omega' \rightarrow \Omega, E' \rightarrow E)$ is used instead of $\Sigma_s(\Omega' \cdot \Omega, E' \rightarrow E)$.
- **Scalar neutron flux** $\psi(\mathbf{x}, E, t)$ is defined as

$$\phi(\mathbf{x}, E, t) = \int_{4\pi} \psi(\mathbf{r}, \Omega, E, t) d\Omega \quad (76)$$

such that $\phi(\mathbf{x}, E, t) dV dE$ means the rate at which path length is generated by neutrons in $dV dE$ about $(\mathbf{x}, E)$ at time t .

- Internal sources

$$\frac{Q(\mathbf{x}, E, t)}{4\pi} \rightarrow S(\mathbf{x}, E, t) \quad \text{or} \quad s(\mathbf{x}, E, t) \quad (77)$$

- The unit of solid angle is **steradian**. The steradian was formerly an SI supplementary unit, but this category was abolished in 1995 and the steradian is now considered an SI derived unit. It is equivalent to $1\text{cm}^2/\text{cm}^2$.